Exact paper stack: original (A) vs math-layout (D)

bayes.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$p(\theta \mid \mathscr{D}) = \frac{p(\mathscr{D} \mid \theta)\, p(\theta)}{\int_\Theta p(\mathscr{D} \mid \vartheta)\, p(\vartheta)\, \mathrm{d}\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta)\, p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)}\left[\prod_{i=1}^n p(y_i \mid x_i, \vartheta)\right]} + \frac{1}{2}\log\left(\frac{|\Sigma_0|}{\left|\Sigma_{\theta|\mathscr{D}}\right|}\right) + \mathrm{KL}\left(q(\theta)\,\|\,p(\theta)\right) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta$$

D: math-layout

$$\begin{aligned} p(\theta \mid \mathscr{D}) &= \frac{p(\mathscr{D} \mid \theta)\, p(\theta)}{\int_\Theta p(\mathscr{D} \mid \vartheta)\, p(\vartheta)\, \mathrm{d}\vartheta} = \frac{\prod_{i=1}^n p(y_i \mid x_i, \theta)\, p(\theta)}{\mathbb{E}_{\vartheta \sim p(\vartheta)}\left[\prod_{i=1}^n p(y_i \mid x_i, \vartheta)\right]} \\ &\quad + \frac{1}{2}\log\left(\frac{|\Sigma_0|}{\left|\Sigma_{\theta|\mathscr{D}}\right|}\right) + \mathrm{KL}\left(q(\theta)\,\|\,p(\theta)\right) + \sum_{i=1}^n \log p(y_i \mid x_i, \theta) - \frac{1}{2}\log\det\Sigma_{\theta|\mathscr{D}} \end{aligned}$$

diffusion.tex eq.0 (continuation, 2 lines)

A: original

$$\mathbb{E}\left[\|x_{t+1} - x_t\|^2\right] = \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E}\left[\|\epsilon_\theta(x_t, t) - \epsilon\|^2\right] + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2}\|\nabla_x \log p_t(x_t) - s_\theta(x_t, t)\|^2$$

D: math-layout

$$\begin{aligned} \mathbb{E}\left[\|x_{t+1} - x_t\|^2\right] &= \sqrt{\frac{2\bar{\alpha}_t}{1 - \bar{\alpha}_t}} \cdot \mathbb{E}\left[\|\epsilon_\theta(x_t, t) - \epsilon\|^2\right] \\ &\quad + \beta_t \|\nabla_x \log p_t(x_t)\|^2 + \frac{\beta_t}{2}\|\nabla_x \log p_t(x_t) - s_\theta(x_t, t)\|^2 \end{aligned}$$

esr.tex eq.0 (continuation, 2 lines)

A: original

$$\mathscr{L}(\theta) = \frac{1}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2\sigma^2}\sum_{i=1}^n \left(y_i - f_\theta(x_i)\right)^2 + \frac{1}{2}\log|\Sigma_\theta| + \frac{d}{2}\log(2\pi) + \lambda\sum_{j=1}^p \left|\theta_j\right| + \frac{n}{2}\log\left(\frac{1}{n}\sum_{i=1}^n (y_i - \hat{y}_i)^2\right)$$

D: math-layout

$$\begin{aligned} \mathscr{L}(\theta) &= \frac{1}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2\sigma^2}\sum_{i=1}^n \left(y_i - f_\theta(x_i)\right)^2 + \frac{1}{2}\log|\Sigma_\theta| \\ &\quad + \frac{d}{2}\log(2\pi) + \lambda\sum_{j=1}^p \left|\theta_j\right| + \frac{n}{2}\log\left(\frac{1}{n}\sum_{i=1}^n (y_i - \hat{y}_i)^2\right) \end{aligned}$$

kalman.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$P_{t|t} = P_{t|t-1} - P_{t|t-1}H_t^\top \left(H_t P_{t|t-1} H_t^\top + R_t\right)^{-1} H_t P_{t|t-1} + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t$$

D: math-layout

$$\begin{aligned} P_{t|t} &= P_{t|t-1} - P_{t|t-1} H_t^\top \left(H_t P_{t|t-1} H_t^\top + R_t \right)^{-1} H_t P_{t|t-1} \\ &\quad + F_t P_{t-1|t-1} F_t^\top - K_t S_t K_t^\top + \Gamma_t u_t u_t^\top \Gamma_t^\top - \Phi_t \Sigma_{t|t-1} \Phi_t^\top + \Omega_t \end{aligned}$$

kkt.tex eq.0 (continuation, 2 lines)

A: original

$$\mathscr{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m \left(\max\{0, g_i(x)\} \right)^2 + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^\star\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho}$$

D: math-layout

$$\begin{aligned} \mathscr{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \nu_j h_j(x) + \frac{\rho}{2} \sum_{i=1}^m \left(\max\{0, g_i(x)\} \right)^2 \\ + \frac{\rho}{2} \sum_{j=1}^p h_j(x)^2 + \frac{\mu}{2} \|x - x^\star\|^2 + \sum_{i=1}^m \frac{\lambda_i^2}{2\rho} + \sum_{j=1}^p \frac{\nu_j^2}{2\rho} \end{aligned}$$

koopman.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$\tilde{A} = \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y = \left(\Psi_X^\top \Psi_X \right)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon \left(\Psi_X^\top \Psi_X + \lambda I \right)^{-1} \Psi_X^\top \Psi_Y + \left\| \Psi_Y - \Psi_X \tilde{A} \right\|_F^2 + \lambda \left\| \tilde{A} \right\|_* + \left\| \Psi_X^\top \Psi$$

D: math-layout

$$\begin{aligned} \tilde{A} &= \tilde{U} \tilde{\Sigma} \tilde{V}^\top = \sum_{k=1}^r \tilde{\sigma}_k \tilde{u}_k \tilde{v}_k^\top = \Psi_X^\dagger \Psi_Y \\ &= \left(\Psi_X^\top \Psi_X \right)^{-1} \Psi_X^\top \Psi_Y + \epsilon I + \epsilon \left(\Psi_X^\top \Psi_X + \lambda I \right)^{-1} \Psi_X^\top \Psi_Y + \left\| \Psi_Y - \Psi_X \tilde{A} \right\|_F^2 + \lambda \left\| \tilde{A} \right\|_* + \left\| \Psi_X^\top \Psi_X - I \right\|_F \end{aligned}$$

lyapunov.tex eq.0 (continuation, 2 lines)

A: original

$$\lambda_{\max} \left(A^\top P + PA + Q \right) \leq -\gamma \|x(t)\|^2 + \limsup_{T \to \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, \mathrm{d}t - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, \mathrm{d}s - \delta \|x\| \|\dot{x}\|$$

D: math-layout

$$\begin{aligned} \lambda_{\max} \left(A^\top P + PA + Q \right) \leq -\gamma \|x(t)\|^2 + \limsup_{T \to \infty} \frac{1}{T} \int_0^T \log \|\Phi(t, t_0)\| \, \mathrm{d}t \\ - \varepsilon \frac{\|x\|^4}{1 + \|x\|^2} + \int_0^t \|\nabla V(x(s))\|^2 \, \mathrm{d}s - \delta \|x\| \|\dot{x}\| \end{aligned}$$

matrix.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$P = A\left(A^\top A + \lambda I\right)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} = U_r U_r^\top - \left(I - U_r U_r^\top\right) + 2 U_r U_r^\top - \left(I - U_r U_r^\top\right) + \sum_{k=1}^{r} \sigma_k u_k^\top v_k - \left(I - U_r U_r^\top\right) A \left(A^\top A + \lambda\right.$$

D: math-layout

$$\begin{aligned} P &= A\left(A^\top A + \lambda I\right)^{-1} A^\top = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} \\ &= U_r U_r^\top - \left(I - U_r U_r^\top\right) + 2 U_r U_r^\top - \left(I - U_r U_r^\top\right) + \sum_{k=1}^{r} \sigma_k u_k^\top v_k - \left(I - U_r U_r^\top\right) A \left(A^\top A + \lambda I\right)^{-1} \end{aligned}$$

matrix.tex eq.1 (continuation-unanchored, 2 lines)

A: original

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \qquad A^\dagger = V_r \Sigma_r^{-1} U_r^\top = \sum_{k=1}^{r} \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^{r} \sigma_k u_k^\top v_k - \left(I - U_r U_r^\top\right) A \left(A^\top A + \lambda I\right)^{-1} + \lambda \mathcal{N}(A)^\perp$$

D: math-layout

$$\begin{aligned} \mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \qquad A^\dagger &= V_r \Sigma_r^{-1} U_r^\top \\ &= \sum_{k=1}^{r} \frac{1}{\sigma_k} v_k u_k^\top + \sum_{k=1}^{r} \sigma_k u_k^\top v_k - \left(I - U_r U_r^\top\right) A \left(A^\top A + \lambda I\right)^{-1} + \lambda \mathcal{N}(A)^\perp \end{aligned}$$

nash.tex eq.0 (continuation, 2 lines)

A: original

$$\sigma^\star \in \arg\max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j + \eta \sum_{i \in S} \sum_{j \in T} \left(\sigma_i \tau_j\right)^2 - \kappa \left\|\sigma - \sigma^\star\right\|_1 + \gamma \left\|\nabla_\sigma \mathbb{E}\left[u\right]\right\|^2 - \beta \sum_{i \in S}$$

D: math-layout

$$\begin{aligned} \sigma^\star \in \arg\max_{\sigma \in \Delta} \min_{\tau \in \Delta} \sum_{i \in S} \sum_{j \in T} \sigma_i \tau_j u_{ij} + \lambda \sum_{i \in S} \sigma_i \log \sigma_i - \mu \sum_{j \in T} \tau_j \log \tau_j \\ + \eta \sum_{i \in S} \sum_{j \in T} \left(\sigma_i \tau_j\right)^2 - \kappa \left\|\sigma - \sigma^\star\right\|_1 + \gamma \left\|\nabla_\sigma \mathbb{E}\left[u\right]\right\|^2 - \beta \sum_{i \in S} \sigma_i^2 \end{aligned}$$

network-flow.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$\sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \qquad \forall v \in V, \qquad 0 \le x_a \le u_a, \qquad \forall a \in A, \qquad \sum_{v \in V} b_v = 0, \qquad \sum_{a \in A} c_a x_a \le C_{\max}, \qquad x_a \ge$$

D: math-layout

$$\begin{aligned} \sum_{a \in \delta^+(v)} x_a - \sum_{a \in \delta^-(v)} x_a = b_v, \qquad \forall v \in V, \qquad 0 \le x_a \le u_a, \qquad \forall a \\ \in A, \qquad \sum_{v \in V} b_v = 0, \qquad \sum_{a \in A} c_a x_a \le C_{\max}, \qquad x_a \ge 0 \end{aligned}$$

pathological.tex eq.1 (single-line, 1 lines)

A: original

$$f\left(g\left(h\left(k\left(x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7\right)\right)\right)\right) + \operatorname{MyLongOperatorName}\left(x_1, x_2, x_3\right) = 0$$

pathological.tex eq.2 (continuation, 3 lines)

A: original

$$S = a_1x_1+a_2x_2+a_3x_3+a_4x_4+a_5x_5+a_6x_6+a_7x_7+a_8x_8+a_9x_9+a_{10}x_{10}+a_{11}x_{11}+a_{12}x_{12}+a_{13}x_{13}+a_{14}x_{14}+a_{15}x_{15}+a_{16}x_{16}+a$$

D: math-layout

$$\begin{aligned} S = {} & a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 + a_5x_5 + a_6x_6 + a_7x_7 + a_8x_8 + a_9x_9 + a_{10}x_{10} \\ & + a_{11}x_{11} + a_{12}x_{12} + a_{13}x_{13} + a_{14}x_{14} + a_{15}x_{15} + a_{16}x_{16} + a_{17}x_{17} \\ & + a_{18}x_{18} + a_{19}x_{19} + a_{20}x_{20} + a_{21}x_{21} + a_{22}x_{22} + a_{23}x_{23} + a_{24}x_{24} \end{aligned}$$

pathological.tex eq.4 (continuation-unanchored, 2 lines)

A: original

$$\operatorname{MyLongOperatorName}\left(\operatorname{MyLongOperatorName}\left(a_1 + a_2 + a_3 + a_4\right) + \left(b_1 + b_2 + b_3 + b_4\right)\right) = \left[c_1 + c_2 + c_3 + c_4 + c_5\right]$$

D: math-layout

$$\begin{aligned} & \operatorname{MyLongOperatorName}\left(\operatorname{MyLongOperatorName}\left(a_1 + a_2 + a_3 + a_4\right) + \left(b_1 + b_2 + b_3 + b_4\right)\right) \\ & \quad = \left[c_1 + c_2 + c_3 + c_4 + c_5\right] + \left\{d_1 + d_2 + d_3\right\} \end{aligned}$$

pathological.tex eq.8 (continuation-unanchored, 2 lines)

A: original

$$(\alpha_1x_1+\alpha_2x_2+\alpha_3x_3+\alpha_4x_4+\alpha_5x_5+\alpha_6x_6+\alpha_7x_7+\alpha_8x_8+\alpha_9x_9+\alpha_{10}x_{10}+\alpha_{11}x_{11}+\alpha_{12}x_{12}+\alpha_{13}x_{13}+\alpha_{14}x_{14}+\alpha_{15}x_{15}+\alpha_{16}x_{16}+$$

D: math-layout

$$\begin{aligned} & (\alpha_1x_1 + \alpha_2x_2 + \alpha_3x_3 + \alpha_4x_4 + \alpha_5x_5 + \alpha_6x_6 + \alpha_7x_7 + \alpha_8x_8 + \alpha_9x_9 + \alpha_{10}x_{10} \\ & \quad + \alpha_{11}x_{11} + \alpha_{12}x_{12} + \alpha_{13}x_{13} + \alpha_{14}x_{14} + \alpha_{15}x_{15} + \alpha_{16}x_{16} + \alpha_{17}x_{17} + \alpha_{18}x_{18}) = 1 \end{aligned}$$

residual-xgb.tex eq.0 (continuation-unanchored, 2 lines)

A: original

$$G(r) = L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2 - \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}_{\text{XGB}}(x_i)\right)^2 \geq \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}_{\text{lin}}(x_i)\right)^2 + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i)\right)^2$$

D: math-layout

$$
\begin{aligned}
G(r) &= L_{\text{null}} - L_{\text{XGB}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2 - \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}_{\text{XGB}}(x_i)\right)^2 \\
&\geq \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}_{\text{lin}}(x_i)\right)^2 + \frac{1}{n}\sum_{i=1}^{n}\left(\hat{f}_{\text{lin}}(x_i) - \hat{f}_{\text{XGB}}(x_i)\right)^2
\end{aligned}
$$